Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
MAKEUP EXAMINATIONS, DECEMBER 2022

DATA STRUCTURES & ALGORITHMS [ELE 4078]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 29/12/2022** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Describe the asymptotic notations associated with algorithms ***(04)***

1B. Solve using Dynamic programming
C= 100; W=[25,35,40,60]; P= [10,12,14,20] ***(04)***

1C. Summarize the applications of stack. ***(02)***

2A. Classify the user-defined data structures. Give example for each. ***(04)***

2B. Solve the recurrence equation using substitution method and determine the time complexity.

$T(n) = d$ if $n=1$;

$= T(n/2) + C$ if $n>1$ ***(04)***

2C. What are the disadvantages of Queue? How is it overcome? ***(02)***

3A. How many pointers are necessary to implement a simple Linked List? Discuss each. ***(03)***

3B. Construct a binary tree using the following details.
In order [ ] ={ D B E A F C G }; Post order[ ] = { D E B F G C A } ***(03)***

3C. Develop a pseudocode to develop a doubly linked list. How do you delete the last node in doubly linked lis. ***(04)***

4A. Write pseudocode/algorithm to PUSH and POP elements to/from a STACK. ***(04)***

4B. Write a pseudocode/algorithm to print the sorted list of elements contained in a Binary search tree. ***(02)***

4C. Convert the following string into its postfix expression.
(300+23)*(43-21)/(84+7) ***(04)***

5A. Write a note on hashing with an example. ***(04)***

5B. Write a pseudocode to initialize an array with natural numbers. Write suitable comments. ***(04)***

5C. Which type of memory allocation is referred for Linked List? Why? ***(02)***